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END SEMESTER PROJECT

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ABSTRACT

This project focuses on the development of an efficient, GUI-based database management system for handling laboratory equipment issuance in our department. Currently, lab attendants manually log equipment lending in physical registers, which is prone to human error and data loss. Our solution offers a role-based digital system built using the .NET Framework and Microsoft SQL Server. The application allows users to register as students, lab attendants, or faculty members. For now, lab attendants have full functionality to manage lab inventory, including issuing components such as microprocessors, LCDs, I2C chips, light bulbs, and breadboards. Key features include secure login/signup, component issuance and return, status tracking, and inventory updates. The system is designed to replace the outdated manual process with a structured, accessible, and secure digital platform.

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INTRODUCTION

Laboratory equipment management is a critical component of engineering departments, especially when dealing with a large number of students and electronic components. Traditional manual logging systems often fall short in terms of reliability, efficiency, and scalability. This project addresses these limitations by providing a comprehensive, desktop-based lab inventory management solution built on the .NET Framework and SQL Server. Through a role-based system, users such as lab attendants can securely handle day-to-day lab operations with improved accuracy and accountability.

Background

Each laboratory in the department operates independently under the supervision of a lab attendant. The attendants maintain a physical register for component issuance, recording only the student's name, registration number, and degree program. Over time, this manual system has proven insufficient due to the growing number of students and the diversity of components. The lack of a unified digital platform prevents easy tracking of issued and returned items, which results in missing or duplicated records, poor inventory visibility, and inefficient lab operations. To address these challenges, a centralized and digitized solution is required.

Problem Statement

The primary issue in the current lab management system is the lack of reliable and organized component tracking. Records are kept in hardcopy notebooks, which makes data retrieval, consistency, and long-term maintenance highly inefficient. This system introduces frequent inconsistencies, redundant entries, and poor traceability. It also creates unplanned dependency on individual lab attendants and offers no backup or real-time status of issued components. If the scale of lab operations increases in the future, these shortcomings could cause critical management failures. Therefore, a digital solution is necessary to maintain consistent and accurate records of lab components, minimize redundancy, and support centralized, scalable operations.

Purpose and Scope of the Project

The purpose of this project is to develop a desktop application that facilitates the management of laboratory equipment issuance using a digital platform. This system will allow lab attendants to securely issue, track, and manage electronic components while maintaining an accurate, centralized record. Built using the .NET Framework and SQL Server, the project includes a role-based login system for lab attendants, with planned future support for students and faculty. The scope of this implementation includes user authentication, component issuance and return, real-time tracking, status updates, and complete inventory control—all accessible through a clean and efficient graphical user interface.

Objectives

- Replace the manual, paper-based tracking system with a digital application.
- Ensure high data consistency and eliminate redundancy using relational database management principles.
- Maintain strong data integrity and accurate records at all times through GUI-backend integration.
- Provide lab attendants with tools to issue, return, add, and remove components securely and efficiently.
- Implement a role-based login and signup system to support controlled access.
- Offer real-time visibility of equipment status and availability.
- Provide a scalable structure that can support future feature expansion for students and faculty users.
- Maintain clear audit trails for equipment usage and ensure traceability of all transactions.

LITERATURE REVIEW

The evolution of inventory and issuance systems within academic institutions has increasingly shifted toward digital platforms due to the inadequacies of manual tracking methods. Traditionally, laboratory equipment was recorded in handwritten registers, leading to inefficiencies in traceability, redundancy in data entry, and lack of real-time access. Bhaskar, Kolhe, and Raut (2023) emphasize that “manual inventory processes are time-consuming and error-prone, making them unsuitable for growing institutions where real-time access and reliability are crucial,” highlighting the growing demand for intelligent inventory systems that are digitally centralized (Bhaskar et al., 2023).

A notable example of this shift can be found in the development of the Lab Inventory Tracking System (LITS) at Universiti Teknologi PETRONAS. This system digitized the lab's inventory records and improved accessibility, reliability, and the generation of real-time reports. Rosli (2015) found that such a platform not only facilitated efficient lab operations but also reduced the dependency on individual staff members for record maintenance. Similarly, Deshmukh and Kore (2019) implemented an MS Access-based inventory control system for an analytical development laboratory, reporting a significant improvement in component availability and overall record accuracy.

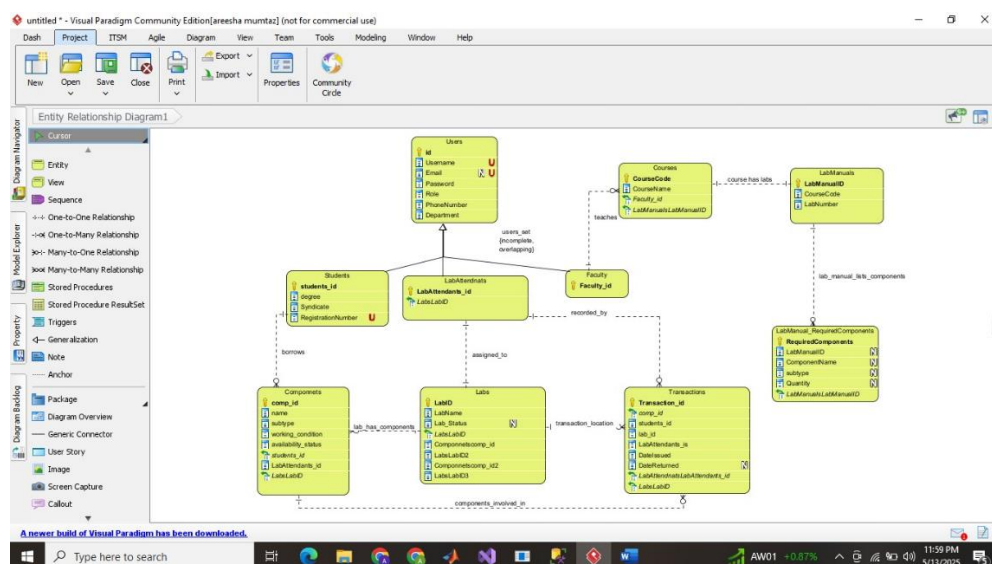
Further innovations have introduced automation and real-time tracking using barcode and QR code technologies. A study by Shelke, Kadam, and Sarode (2021) described a QR-code-based inventory system implemented in a technical institution, where each component was tagged and easily scanned by lab attendants. This system not only reduced human error but also allowed for a smoother user experience. According to their findings, “QR code technology helps create transparency in the inventory, streamlining both issuance and return processes” (Shelke et al., 2021).

Broader implementations of digital inventory systems in universities and research labs also underscore the necessity of centralized data management. For example, Herrera, López, and Rodríguez (2022) designed a low-cost, web-based information system to manage chemical inventory within university departments. Their system ensured role-based access, audit trails, and clear inventory visibility—characteristics that our project similarly aims to emulate. Likewise, Pavlova et al. (2024) introduced a web-based goods and inventory management solution for academic health science laboratories, reporting that such tools reduced time spent on manual stock-taking and allowed faculty to “focus on more intellectually valuable tasks” rather than administrative ones.

These previous works demonstrate not only the feasibility but also the importance of replacing manual systems with digital ones in laboratory environments. The literature makes it clear that implementing a GUI-based database platform provides substantial gains in data consistency, reliability, and operational scalability. These findings heavily informed the direction of this project, which aims to replicate and extend these benefits in our department's laboratory component issuance process.

SYSTEM DESIGN

Entity-Relationship (ER) Diagram



Technologies Used

The following technologies were used in the development of the Lab Inventory Management System:

- **Frontend:** Windows Forms (WinForms) using the **.NET Framework**
- **Backend:** C# with **ADO.NET** for data operations
- **Database:** **Microsoft SQL Server** for relational data storage
- **IDE:** Visual Studio
- **Authentication & Security:** Custom-built login module with role-based access.

Frontend & Backend Development

- The frontend was designed using Windows Forms to create a user-friendly interface for lab attendants, faculty, and students.
- The backend logic was written in C# and connected to the SQL Server database using ADO.NET. All CRUD (Create, Read, Update, Delete) operations are handled via SQL queries or stored procedures.
- Role-based dashboards are dynamically loaded depending on the user's role at login.

Database Implementation

- A relational database schema was designed to support components, users, transaction history, and lab-specific requirements.
- Core tables include:
 - **Users** (stores user credentials and roles)
 - **Components** (details of lab equipment/components)
 - **Transaction_History** (records of issuance and return)
 - **Labs** (lab info and lab attendant assignments)
 - **Lab_Requirements** (required components per lab session)
- Proper foreign key relationships were established to maintain data integrity.

Security Features

- **Role-Based Access Control (RBAC):**
 - Upon login, users are redirected to dashboards based on their roles (Student, Lab Attendant, Faculty).
 - Lab attendants can only manage the labs assigned to them.
- **Authentication Validation:**
 - Unique constraints on email and username during registration.
 - Password masking and secure validation during login.
- **Controlled Component Operations:**
 - Only authorized lab attendants can issue, return, or delete components.
 - Admin-level access (if implemented) could be used for reporting and lab reassignments.

SYSTEM FEATURES

Login /Signup Page :

- Frontend Functionality:

The application begins with a Login and Sign-Up page. New users can register by entering the following information:

- Full Name
- Email (must be unique)
- Username (must be unique)
- Password
- Role Selection (Student, Lab Attendant, Faculty)
- Department

Depending on the selected role, additional fields may appear:

- If the user selects "Lab Attendant", a dropdown or input field prompts them to select the lab they will supervise.

- Backend Operations:

• Sign-Up:

- The system performs validation to ensure both the username and email are unique using a `SELECT` query.
- Upon successful validation:
 - A new entry is added to the `Users` table with role-based information.
 - If the role is Lab Attendant:
 - The selected lab is checked in the `Labs` table to ensure it is not already assigned.
 - If unassigned, the lab is linked to the newly registered lab attendant (e.g., by setting `AssignedTo` in the `Labs` table).
 - If already assigned, an error message is shown and registration is blocked .

• Login:

- The user enters their credentials.
- A `SELECT` query checks for matching credentials in the `Users` table.
- Upon successful authentication, the user is redirected to a role-specific dashboard.

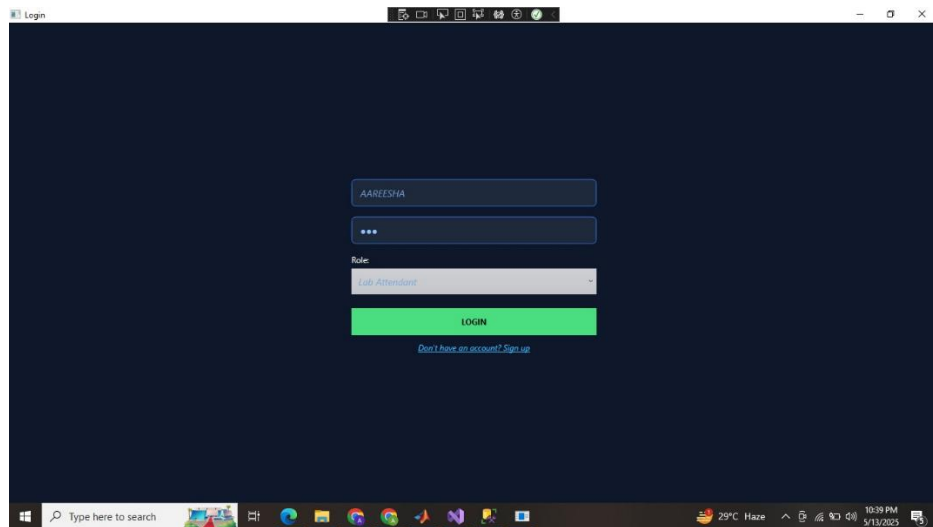


Figure 1: Login Page

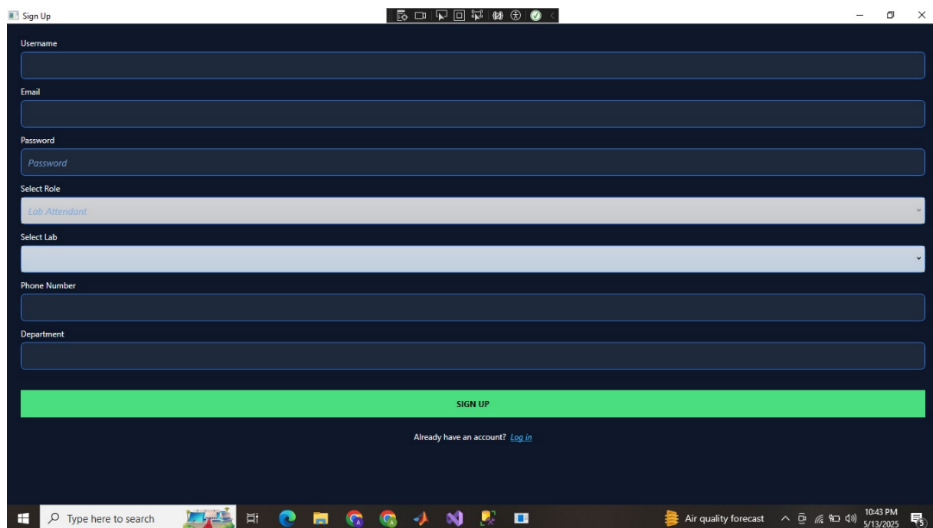


Figure 2: Sign up Page tailored for the role of lab attendant.

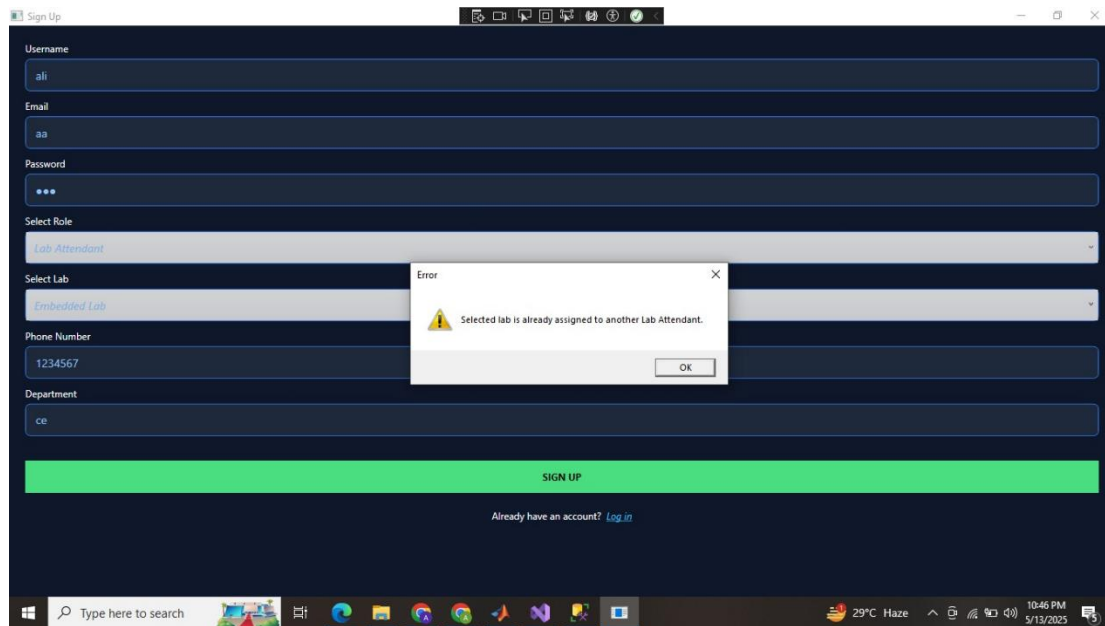


Figure 3 : Error thrown while sign up if lab attendant selects the lab already assigned .

Lab Attendant Dashboard :

- Frontend Functionality:

The Lab Attendant Dashboard provides an intuitive graphical interface for managing laboratory components. Upon logging in, the lab attendant is presented with a centralized control panel featuring multiple operation buttons. Each button corresponds to a specific functionality:

- Component Issuance
- Component Return
- Component Track
- Status Update
- Request Component
- Add Component
- Remove Component
- Exit

When any of these options is selected, the system opens the respective form or module in a new window or section. These forms are designed for:

- Inputting relevant information
- Performing backend updates
- Displaying necessary data from the database

This modular design ensures ease of use, quick access to core operations, and minimizes errors during data handling.

- **Backend Logic:**

Each form is linked to its corresponding database operation (e.g., insert, update, select, or delete queries), and ensures data consistency by performing validation before committing changes.

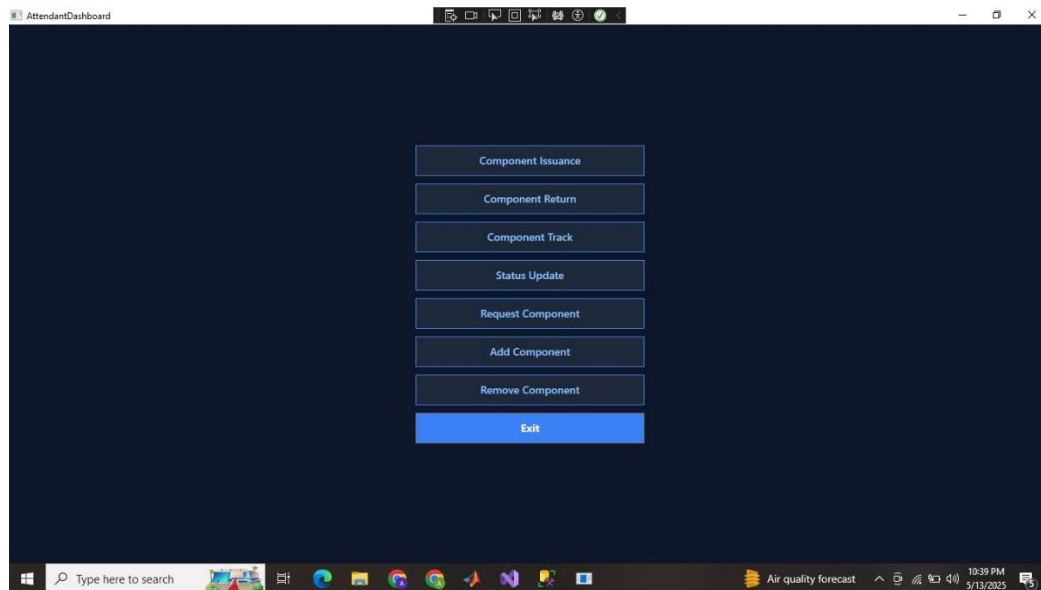


Figure 4: Dashboard for lab attendant.

Component Issuance

- **Frontend Functionality:**

The lab attendant interface provides a button labeled "Component Issuance" (as shown in the UI screenshot). On clicking this, a new form opens where the attendant can search for a component by name or category. Once the desired component is found, the system checks its availability and working status. If the component is available and in working condition, the lab attendant proceeds to issue it by entering the student's details (Name, Registration Number, Degree Program).

- **Backend Operations:**

- A SELECT query is executed to check the availability of the component (Status = 'Available' AND Condition = 'Working').
- If available, a new row is inserted into the Transaction_History table with the following fields:
 - Component ID
 - Student Name
 - Registration Number
 - Degree Program
 - Date and Time of Issuance
 - Issued By (Lab Attendant ID)

- The Status field of the component in the Components table is updated to 'Issued'

The screenshot shows a web application window titled "Issue Component". It features a search bar for "Component Name" containing the text "arduino", followed by a blue button labeled "Search Available Components". Below this is a dropdown menu for "Available Component IDs" which is open, showing a list of IDs: 17, 19, 456, and 789. Under the dropdown are five empty input fields labeled "Student Name", "Registration No.", "Degree", "Department", and "Syndicate". At the bottom of the form is a blue button labeled "Issue Component". The window is set against a dark blue background and is running on a Windows operating system, as indicated by the taskbar at the bottom.

Figure 5: Interface for component issuance.

Component Return

- Frontend Functionality:

When a student returns a component, the lab attendant accesses the "Component Return" form through the dashboard (as shown in the UI). The attendant searches for the component using its unique ID or other identifiers. Once found, the status of the component is reviewed and the status gets automatically updated to 'Available'.

- Backend Operations:

A SELECT query is used to fetch the component record from the Components table using the entered Component ID.

- The Status field of the corresponding record in the Components table is updated from 'Issued' to 'Available'.
- The Transaction_History table is then updated with:
 - The Return Date and Time for the component's transaction entry.

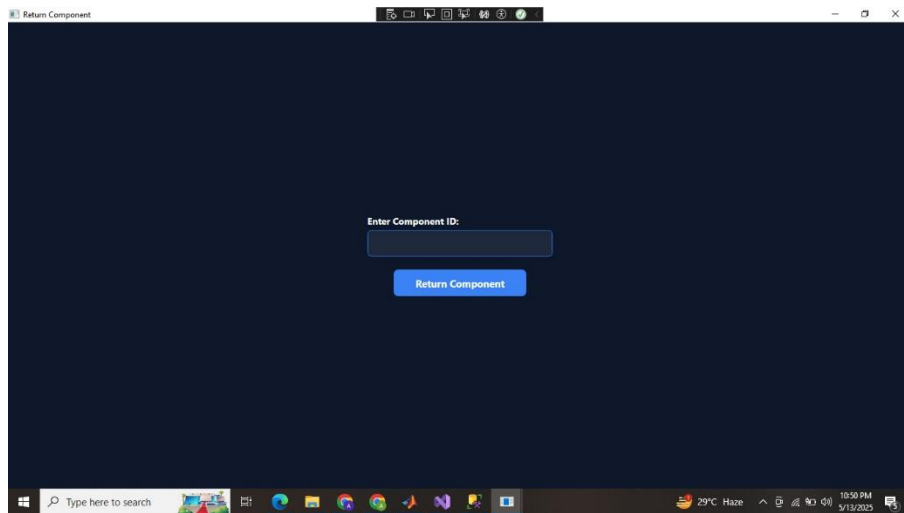


Figure 6: Interface for Component return.

	TransactionId	ComponentId	StudentId	LabId	LabAttendantId	DateIssued	DateReturned	Purpose	Remarks
▶	4	4	3	1	8	2025-09-04	2025-09-05	NULL	NULL
	5	4	3	1	8	2025-05-10	2025-05-10	NULL	NULL
	6	17	3	1	8	2025-05-10	2025-05-10	General Issuance	Issued from das...
	7	20	11	1	8	2025-05-10	2025-05-12	General Issuance	Issued from das...
	8	123	19	1	8	2025-05-12	NULL	General Issuance	Issued from das...
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 7: Component 123 has not been returned yet can be verified from database .

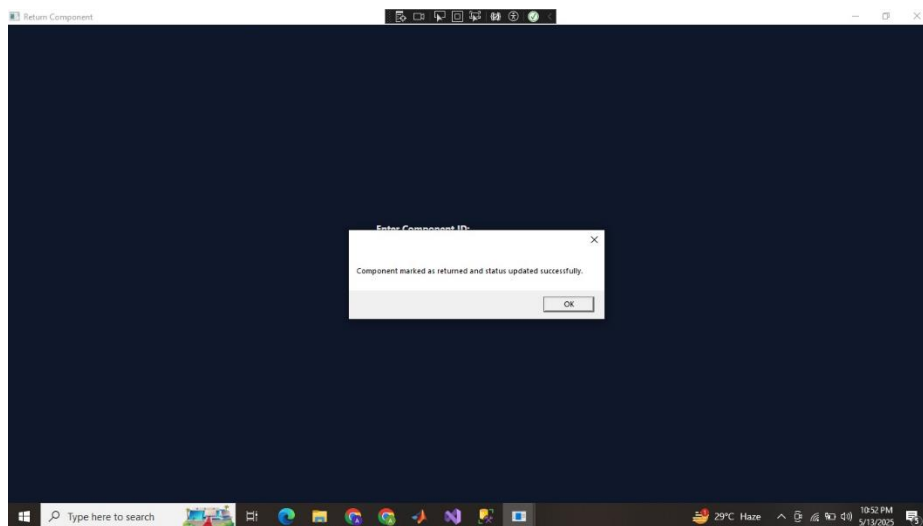


Figure 8: Component 123 returned successfully.

TransactionId	ComponentId	StudentId	LabId	LabAttendantId	DateIssued	DateReturned	Purpose	Remarks
3	4	3	1	8	2025-09-04	2025-09-05	NULL	NULL
5	4	3	1	8	2025-05-10	2025-05-10	NULL	NULL
6	17	3	1	8	2025-05-10	2025-05-10	General Issuance	Issued from des...
7	20	11	1	8	2025-05-10	2025-05-12	General Issuance	Issued from des...
8	123	19	1	8	2025-05-12	2025-05-13	General Issuance	Issued from des...
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 9: Return date of component 123 updated in database.

Remove Component

- Frontend Functionality:

The lab attendant uses the "Remove Component" option to delete components that are no longer usable (e.g., damaged or obsolete). The attendant searches for the component by its unique ID. Once verified, the component is removed from the system.

- Backend Operations:

- A SELECT query retrieves the component details using the provided Component ID for verification.
- Upon confirmation:
 - The component is deleted from the Components table.
 - All associated records of that component in the Transaction_History table are also deleted, ensuring no orphaned historical data remains.

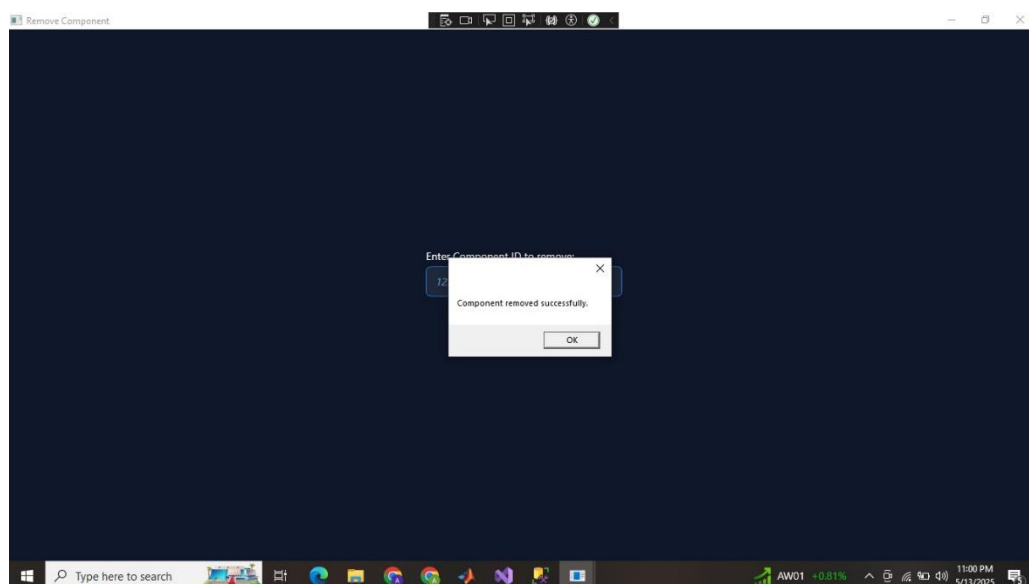


Figure 10: Component 123 removed from database successfully via entering component id.

TransactionId	ComponentId	StudentId	LabId	LabAttendantId	DateIssued	DateReturned	Purpose	Remarks
4	4	3	1	8	2025-09-04	2025-09-05	NULL	NULL
5	4	3	1	8	2025-05-10	2025-05-10	NULL	NULL
6	17	3	1	8	2025-05-10	2025-05-10	General Issuance	Issued from das...
7	20	11	1	8	2025-05-10	2025-05-12	General Issuance	Issued from das...
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 11: Transaction history of component 123 removed from database .

Status Update

The "Status Update" feature allows the lab attendant to update the working condition of a component. For example, if a previously faulty component has been repaired and is now functional, the attendant can search for it using its Component ID and update its condition accordingly.

- Backend Operations:

- A SELECT query fetches the component using the entered Component ID for verification.
- Upon user input, the Condition field of the component in the Components table is updated to reflect the new status (e.g., 'Working', 'Faulty', 'Under Maintenance').

Figure 12 : shows current status

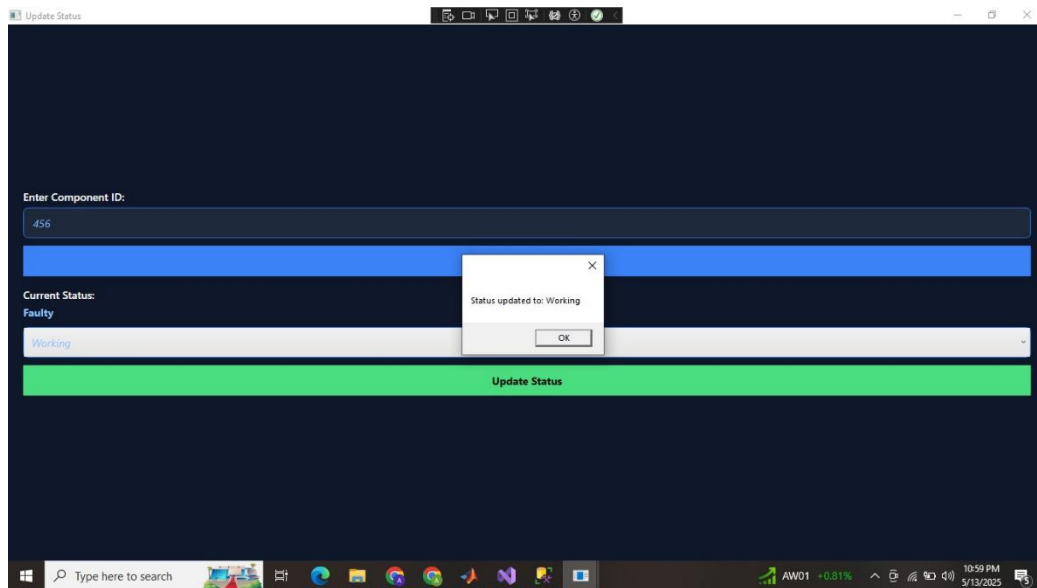


Figure 13: Status of component 456 updated to working .

Component Tracking

- Frontend Functionality:

The "Component Track" feature provides the lab attendant with multiple ways to monitor and manage inventory. It includes three key options:

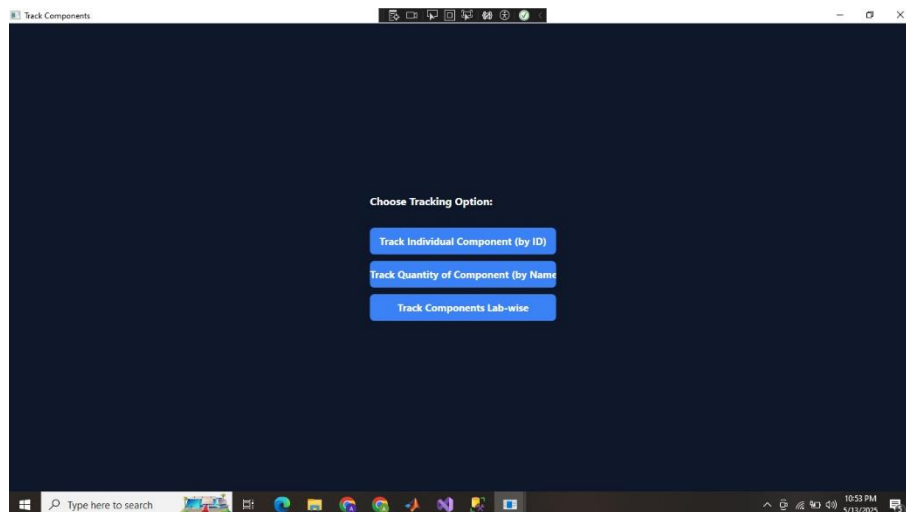


Figure 14: Component Tracking interface.

1. Track by Component ID

The attendant enters the Component ID to retrieve detailed information about a specific component, including its name, status, condition, issue history, and current holder (if issued).

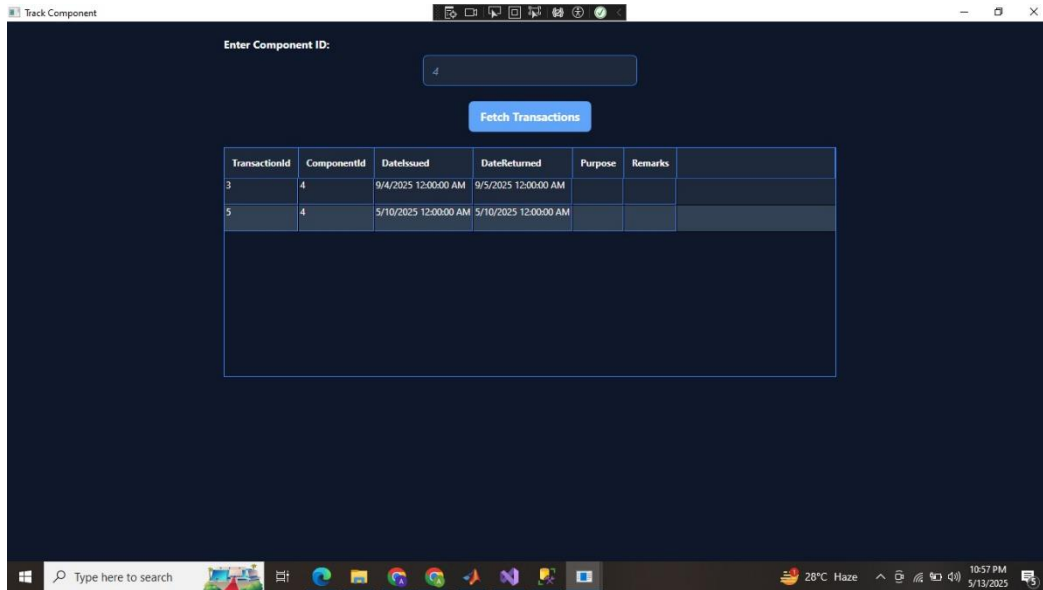


Figure 15: Tracking a component by its id returns its transaction history .

2. Search Quantity by Name or Subtype

The attendant can enter the component name or subtype to view the total quantity available, number issued, and number under maintenance or faulty. This helps in planning and restocking.

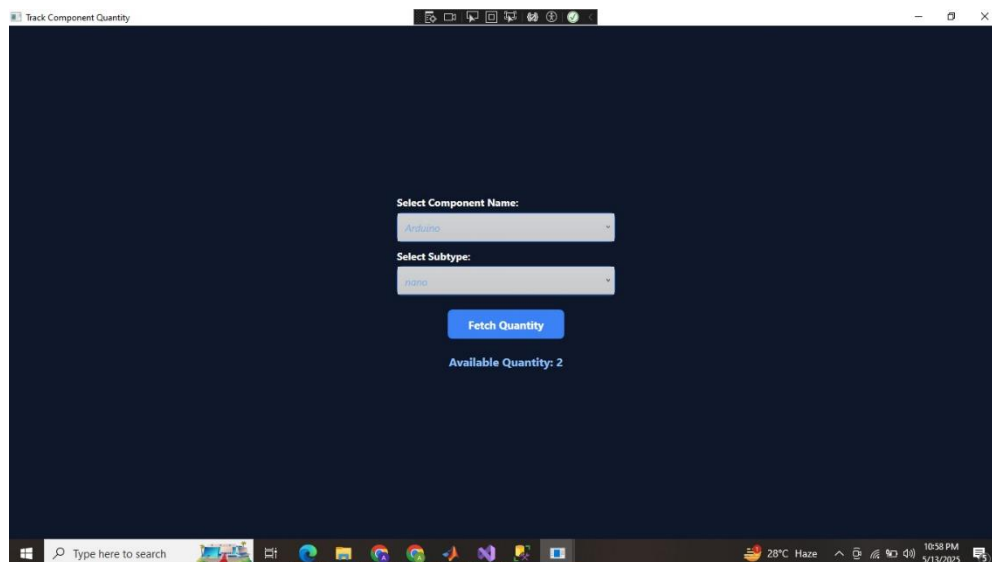


Figure 16: Tracking a component by its name or subtype returns the quantity of respective componnt available.

3. Lab-Specific Component Requirements

The attendant selects a course and lab number to view the list of components required for that lab. A comparison table is displayed with:

- i. Required Quantity (based on the course and lab)
- ii. Available Quantity (currently in inventory)

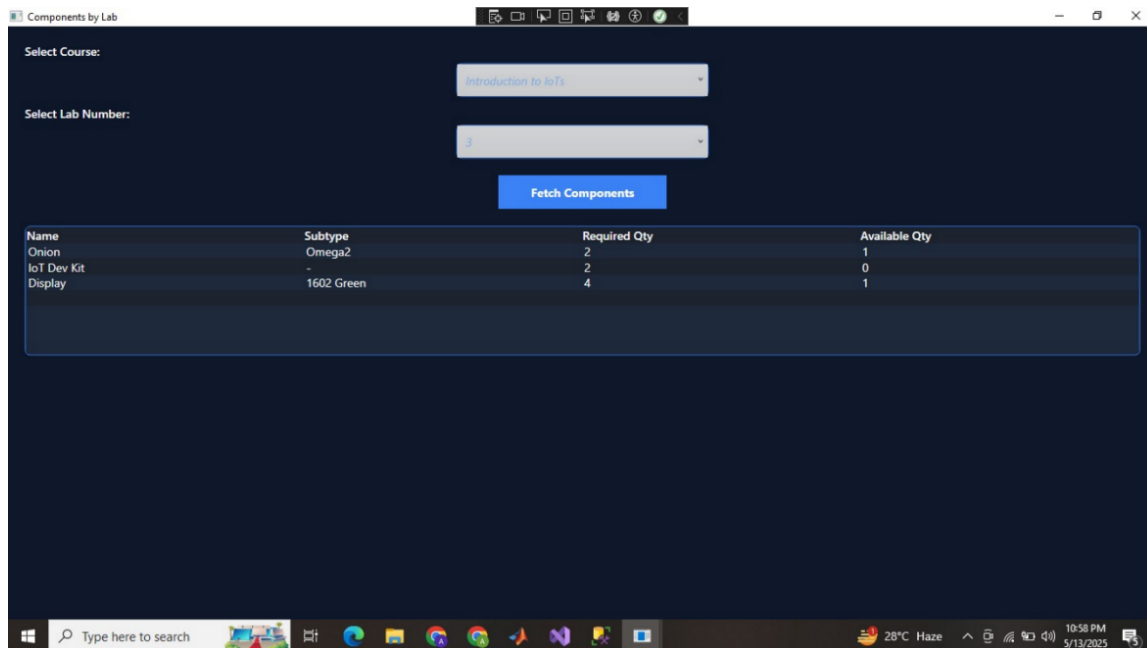


Figure 17: Component required by a certain lab of a certain course can be fetched via entering course code and respective lab number .

- Backend Operations:

- For tracking by ID: a SELECT query fetches full details from the Components and Transaction_History tables.
- For quantity search: a GROUP BY or COUNT query retrieves component counts filtered by name or subtype.
- For lab-specific requirement tracking:
 - The system queries the Lab_Requirements table to retrieve components and required quantities.
 - A JOIN is performed with the Components table to display available vs. required data in a tabular format.

Add Component

- Frontend Functionality:

When new components are delivered to the lab, the lab attendant uses the "Add Component" option from the dashboard. The system first prompts the attendant to enter:

- Type (e.g., Sensor, Resistor, Microcontroller)

- Subtype (e.g., IR Sensor, 10k Ω Resistor, Arduino Uno)
- Quantity (number of units received)

After specifying these details, the system allows the lab attendant to manually enter unique IDs for each component unit. This ensures each item can be individually tracked later.

- **Backend Operations:**

- For each entered ID, a new row is **inserted into the Components table** with the following fields:
 - Component ID (unique)
 - Type
 - Subtype
 - Status (default set to 'Available')
 - Condition (default set to 'Working')
 - Entry Date/Time

Figure 18: Interface for entering new components in database .

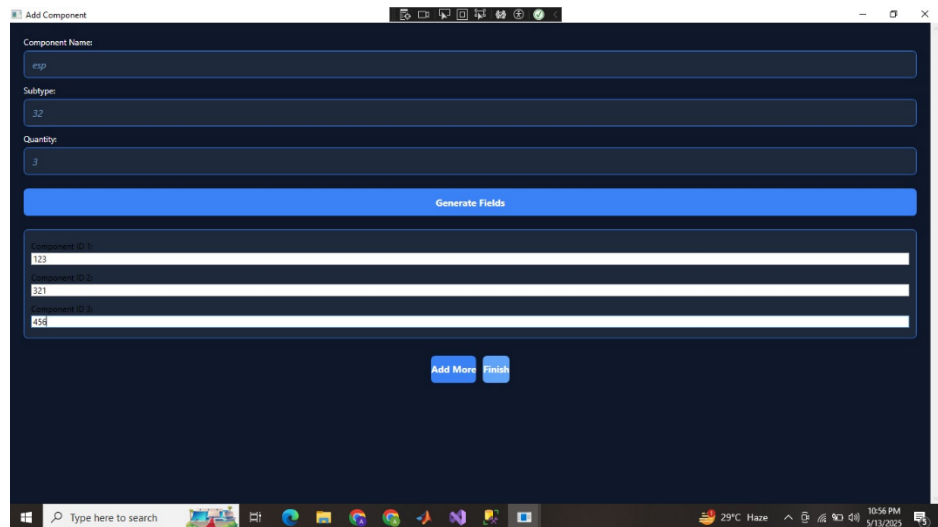


Figure 19: No. of fields generated according to the quantity of new component.

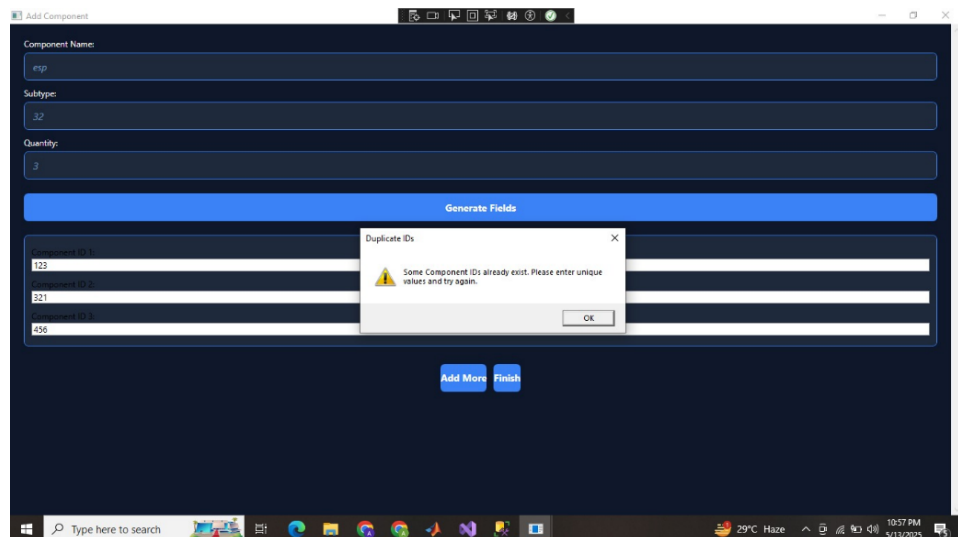


Figure 20: Error thrown if id of new components is not unique .

TESTING AND VALIDATION

Test Cases

A range of test cases was designed to validate the correctness and stability of key features in the system. Some examples include:

Test Case	Description	Expected Result	Status
Login with correct credentials	Validate login functionality	Redirect to role-based dashboard	Passed
Duplicate email on sign-up	Check uniqueness constraint	Show error message	Passed

Component issuance with insufficient quantity	Try issuing a component not in stock	Show "Not available" alert	Passed
Return component	Change status to "Available" and log return time	Status updated, return logged	Passed
Delete component	Remove component and its history	Component deleted from DB	Passed

Testing Strategy

1. Unit Testing:

Each module (login, issuance, return, add component, delete component) was tested individually to verify correct input handling and data manipulation logic.

2. Integration Testing:

Modules were tested in combination to ensure smooth interaction. Example: Sign-up → Login → Issue Component → Return → View History.

3. System Testing:

End-to-end testing was done by simulating actual workflows (lab attendants managing components, issuing/returning parts, viewing status reports).

Tools Used:

- Manual testing via GUI
- SQL queries to verify backend consistency

RESULTS

Issue	Cause	Fix Applied
Component ID duplication	Manual entry without check	Added backend validation to enforce uniqueness
Return time not logging	Date/time field not updating	Corrected SQL update query with current timestamp
Lab assignment conflict	Multiple attendants assigned to same lab	Backend check added to prevent duplicate assignment
Search filters unresponsive	Incorrect SQL LIKE clause	Fixed query logic and UI input handling

Benefits Over the Manual System

The newly implemented digital Lab Inventory Management System offers several clear advantages over the traditional manual record-keeping method:

- **Centralized Data Storage:** All component records, student issuance logs, and lab requirements are now securely stored in a structured database.
- **Improved Accuracy:** Eliminates human errors in handwriting, misplacement of notebooks, and duplicate entries.
- **Real-Time Updates:** Any component issuance, return, or status update reflects immediately in the system.
- **Easy Tracking:** Lab attendants can search and filter data quickly by component ID, name, or course-related lab requirements.
- **Historical Records:** A complete transaction history is maintained, which was not feasible with manual methods.
- **Role-based Access:** Ensures that only authorized personnel perform sensitive operations like deleting components or updating statuses.

Performance Observations

- **Responsiveness:** The application responds quickly to user inputs and queries, even with a moderate-sized database.
- **Scalability:** The database schema is designed to accommodate additional labs, users, and components as the department grows.
- **Data Integrity:** Primary and foreign key constraints, along with input validation at both frontend and backend, ensure consistency and correctness of stored data.
- **System Reliability:** No data loss or crashes were observed during testing under normal usage conditions.

CONCLUSION

Summary of work

This project successfully replaces the traditional manual lab inventory system with a secure, efficient, and user-friendly digital platform. By implementing a role-based desktop application using the .NET Framework and SQL Server, the system improves data accuracy, traceability, and operational efficiency. It streamlines equipment issuance and return processes, reduces human error, and provides a scalable foundation for future enhancements involving students and faculty.

Limitations and Future Enhancements

Currently, the project is designed primarily from the perspective of the lab attendant, focusing on a single lab and tracking components used in two specific courses. While it efficiently manages inventory and logs transactions, it lacks features for student and faculty interaction. In future enhancements, the system can be expanded to allow lab-wise component issuance—not just tracking—enabling more precise resource allocation. Additionally, it can be scaled to support multiple labs, more courses, and integrate user accounts for students and faculty to request or return components directly. Features like real-time stock alerts, QR/barcode scanning for faster logging, and cloud-based access could further improve the system's usability and scope.

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